

Wonwoo Kang

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Research Interest

Cluster Algebra, Representation Theory

Employment

International Center for Mathematical Sciences-Sofia, Bulgaria
Postdoctoral Researcher, June 2025 - present.

Education

University of Illinois Urbana-Champaign, USA
Ph.D., Mathematics, August 2019 - August 2025.
Advisor: Professor Rinat Kedem

Yonsei University, Republic of Korea
B.S., Mathematics, 2014 - 2018.

Preprints

1. **Cluster Expansions from Punctured Orbifolds**
(with E. Banaian, E. Kelley, E. Oğuz, and E. Yıldırım)
Preprint available at <https://arxiv.org/abs/2605.04982> (2026)
2. **Bridging the Know-Act Gap via Task-Level Autoregressive Reasoning**
(with J. Ahn, R. Kamoi, B. Atıl, R. Lou, et al.)
Preprint available at <https://arxiv.org/abs/2603.22619> (2026)
3. **Persistent Laplacian Diagrams**, submitted
(with I. Jung and H. Park)
Preprint available at <https://arxiv.org/abs/2512.05463> (2025)
4. **Skein relations for punctured surfaces**, submitted
(with E. Banaian and E. Kelley)
Preprint available at <https://arxiv.org/abs/2409.04957> (2024)
5. **Accelerated Evaluation of Ollivier-Ricci Curvature Lower Bounds: Bridging Theory and Computation**, submitted
(with H. Park)
Preprint available at <https://arxiv.org/abs/2405.13302> (2024)

Publications	1. Unimodality and Cluster Algebras from Surfaces, submitted (with K. Lee and E. Lim) <i>European Journal of Combinatorics</i>, 136(2026), DOI:10.1016/j.ejc.2026.104392
Proceeding	1. Skein relations for punctured surfaces (with E. Banaian and E. Kelley) <i>Seminaire Lotharingien de Combinatoire in the proceedings for Formal Power Series and Algebraic Combinatorics 2024</i>
Poster	1. Formal Power Series and Algebraic Combinatorics(FPSAC), Ruhr-Universität Bochum, July 2024 <i>Skein relations for punctured surfaces</i> 2. Farey’s legacy in frieze patterns and discrete geometry, ICMS Edinburgh, April 2026 <i>Unimodality from Cluster Algebras from Surfaces</i>
Talks	Skein Relations for Punctured Surfaces DA Seminar, Yonsei University (April 2024) Pusan National University (May 2024) IBS, Daejeon (June 2024) Seminar in Cluster Algebra, Michigan State University (October 2024) UConn Algebra Seminar, University of Connecticut (February 2025) Unimodality and Cluster Algebras from Surfaces 2025 Spring Southeastern AMS Sectional Meeting (March 2025) Combinatorial Expansion Formulae for Generalized Cluster Algebras from Punctured Orbifolds Quantum Groups and Cluster Algebras, International Conference, Bulgaria (December 2025)
Awards and Fellowships	Academic Scholarship Yonsei University, 2014-2015 National Science and Technology Scholarship Korea Student Aid Foundation, 2016-2017 Cryptoanalysis Contest - 2nd place Korea Ministry of National Defense, 2018 R.H. Schark Scholarship University of Illinois, Urbana-Champaign, 2023 Susan C. Morisato IML Scholarship University of Illinois, Urbana-Champaign, 2024

Teaching

Teaching Assistant

University of Illinois Urbana-Champaign

- Spring 2025: Math 285, Intro Differential Equations
- Fall 2024: Math 257, Linear Algebra with Computational Applications
- Spring 2024: Math 257, Linear Algebra with Computational Applications
- Fall 2023: Math 285, Intro Differential Equations
- Spring 2023: Math 241, Calculus II
- Fall 2022: Math 220, Calculus
- Spring 2022: Math 241, Calculus III
- Fall 2021: Math 241, Calculus III
- Spring 2021: Math 234, Calculus for Business I
- Spring 2020: Math 234, Calculus for Business I
- Fall 2019: Math 115, Preparation for Calculus

Yonsei University

- Spring 2019, Calculus 2
- Fall 2018, Calculus 2
- Spring 2018, Calculus 1
- Spring 2018, Discrete Mathematics

Additional Campus Affiliations

- 2024 Spring Illinois Mathematics Lab
- 2023 Fall Illinois Geometry Lab
- 2023-2024 Teaching Award Committee
- 2023-2024 UIUC Association for Women in Mathematics Academic Chair
- 2023 SIM Camp Instructor & Program Coordinator
- 2022-2023 UIUC Association for Women in Mathematics Academic Chair
- 2022-2023 Teaching Award Finalist

Academic Services

Referee for :

- Combinatorial Theory

(Co)-organizer of:

- AWM Graduate Student Colloquium at UIUC
- Student Cluster Algebra Seminar at UIUC

Languages and Skills

Korean (native), English (advanced)
 $\LaTeX$, Mathematica